

Employee Attrition Analysis

IBM HR DATASET · 1,470 EMPLOYEES

CAP 4767 Data Mining Capstone

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The Hidden Cost of Turnover

16%

Annual
Attrition Rate

Nearly 1 in 6 employees
leaves every year

\$5.8M

Replacement Cost
(High)

Upper bound of estimated annual loss

\$3.5M

Replacement Cost
(Low)

Conservative estimate of turnover
expense



Attrition is not uniformly distributed, it is **concentrated in specific roles and conditions**, making it measurable and preventable with the right interventions.



The Dataset & Methodology

IBM HR Analytics Dataset

1,470 employees

35 features · Real-world HR data

Target variable:

Attrition (Yes / No)

Class imbalance addressed:

16% positive class (attrition = Yes)

1

Logistic Regression

Interpretable probability scoring

2

K-Means Clustering

Unsupervised employee segmentation

3

Decision Tree

Plain-English classification rules

The Overtime Crisis

No Overtime

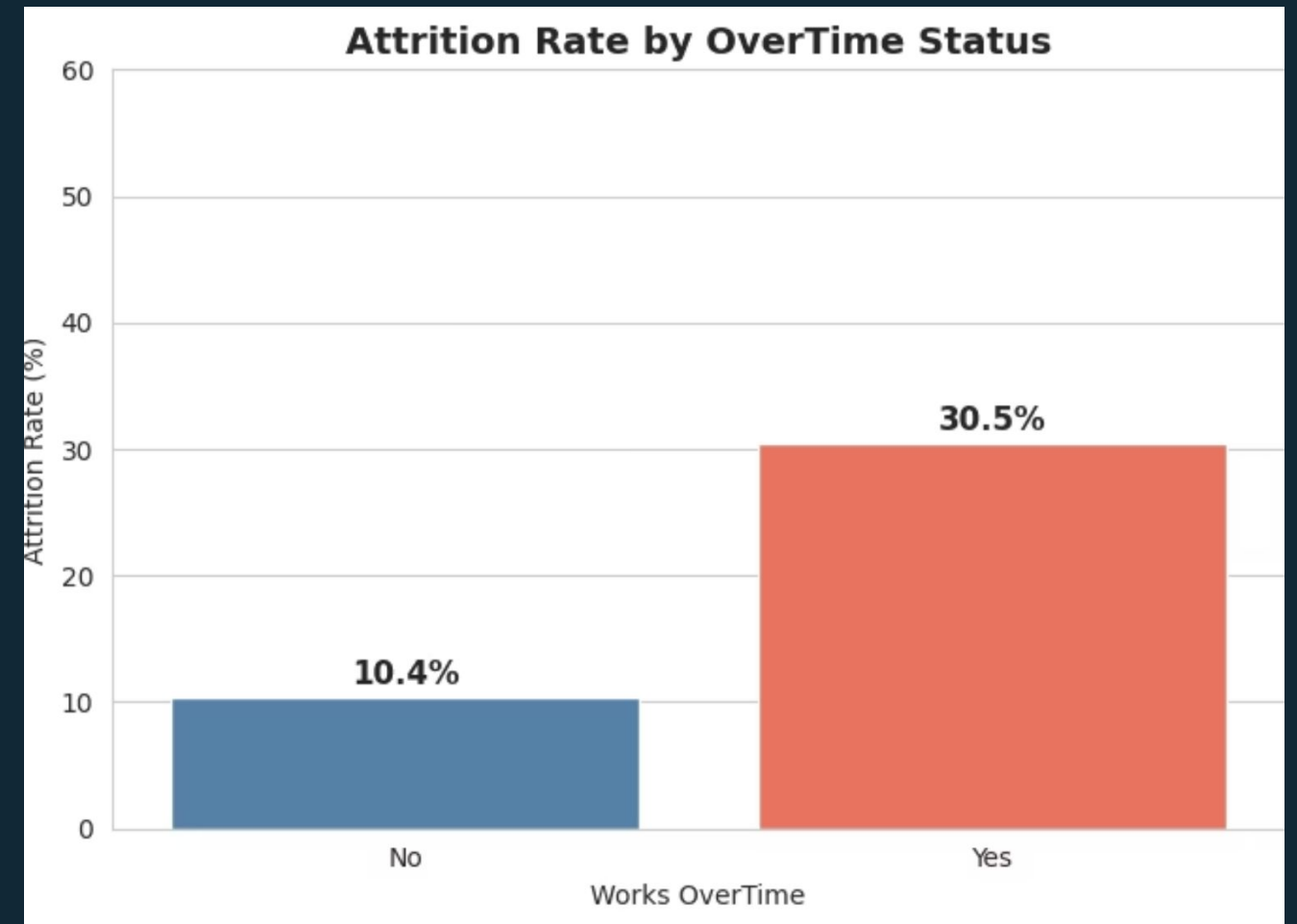
10.4%

Attrition rate for employees with standard hours

Working Overtime

30.5%

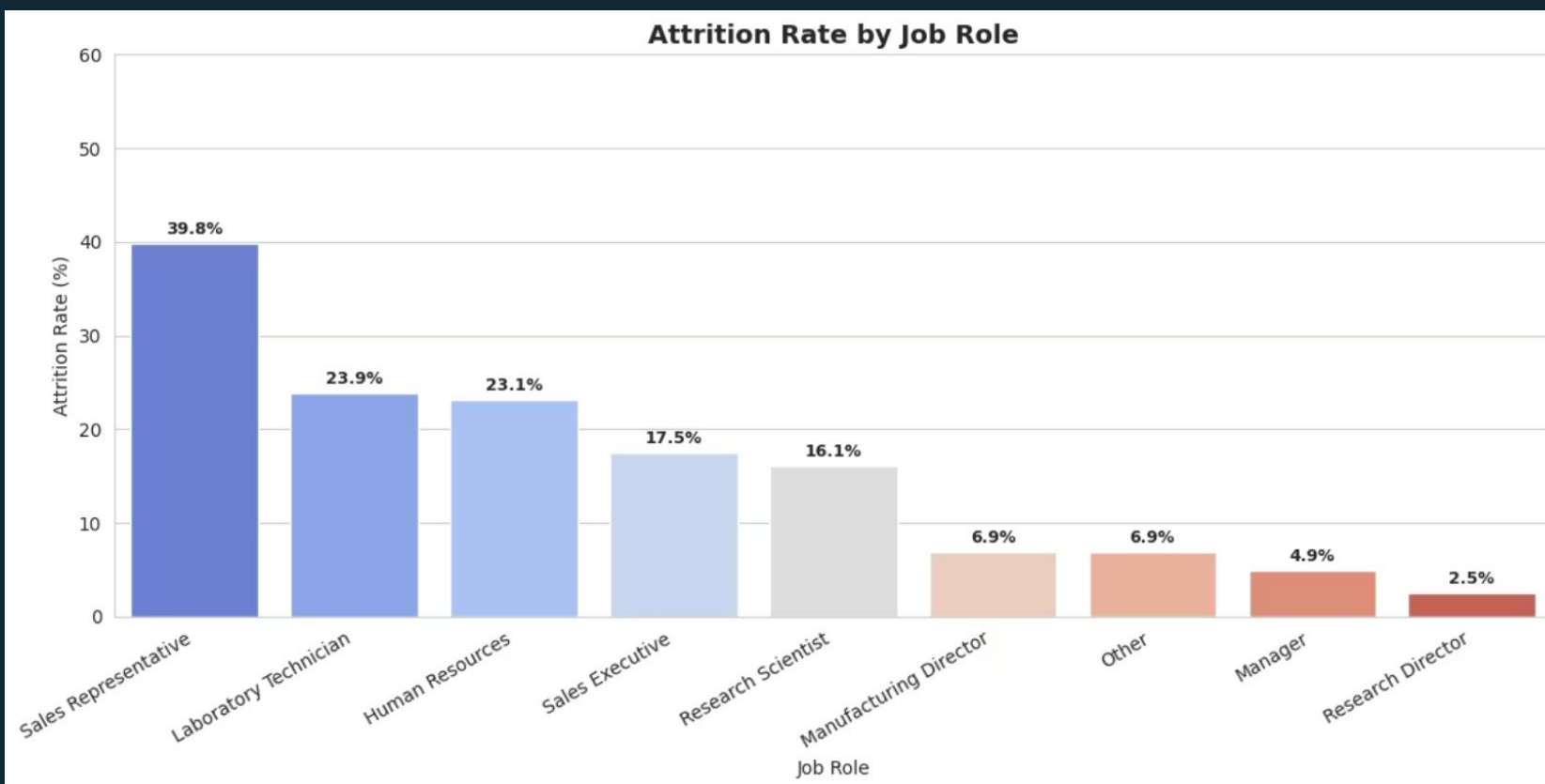
Attrition rate for overtime employees is nearly 3× higher



⚠ Overtime is the **single most controllable factor** in the entire dataset. No other variable shows this magnitude of effect with this level of managerial influence.

FINDING 2

A Role-Specific Crisis, Not a Company-Wide One



Key Insight

Attrition risk is highly concentrated. Sales Representatives leave at a rate **nearly 16× higher** than Research Directors.

Laboratory Technicians also show critically elevated turnover at **23.9%**, a signal of structural role-level stress.

Targeted role-specific interventions will be far more cost-effective than company-wide programs.

FINDING 3

Three Distinct Employee Segments

 **Established Veterans**

8.0% attrition rate

Most stable group. Strong tenure, compensation, and role satisfaction.

 **Early Career**

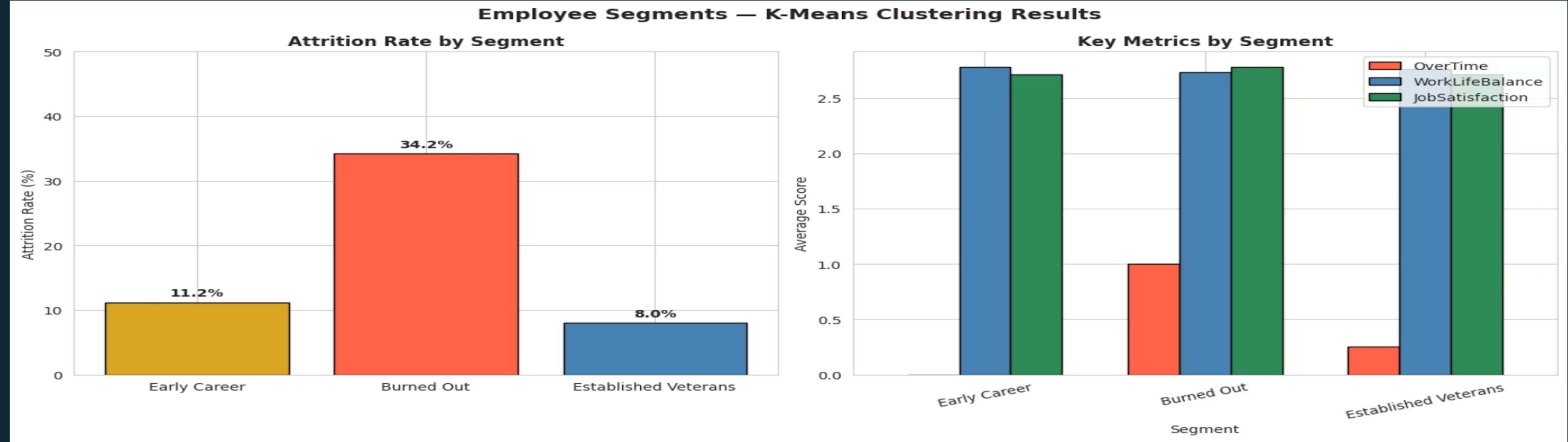
11.2% attrition rate

Younger employees seeking growth and advancement. Retention tied to visible promotion pathways.

 **Burned Out**

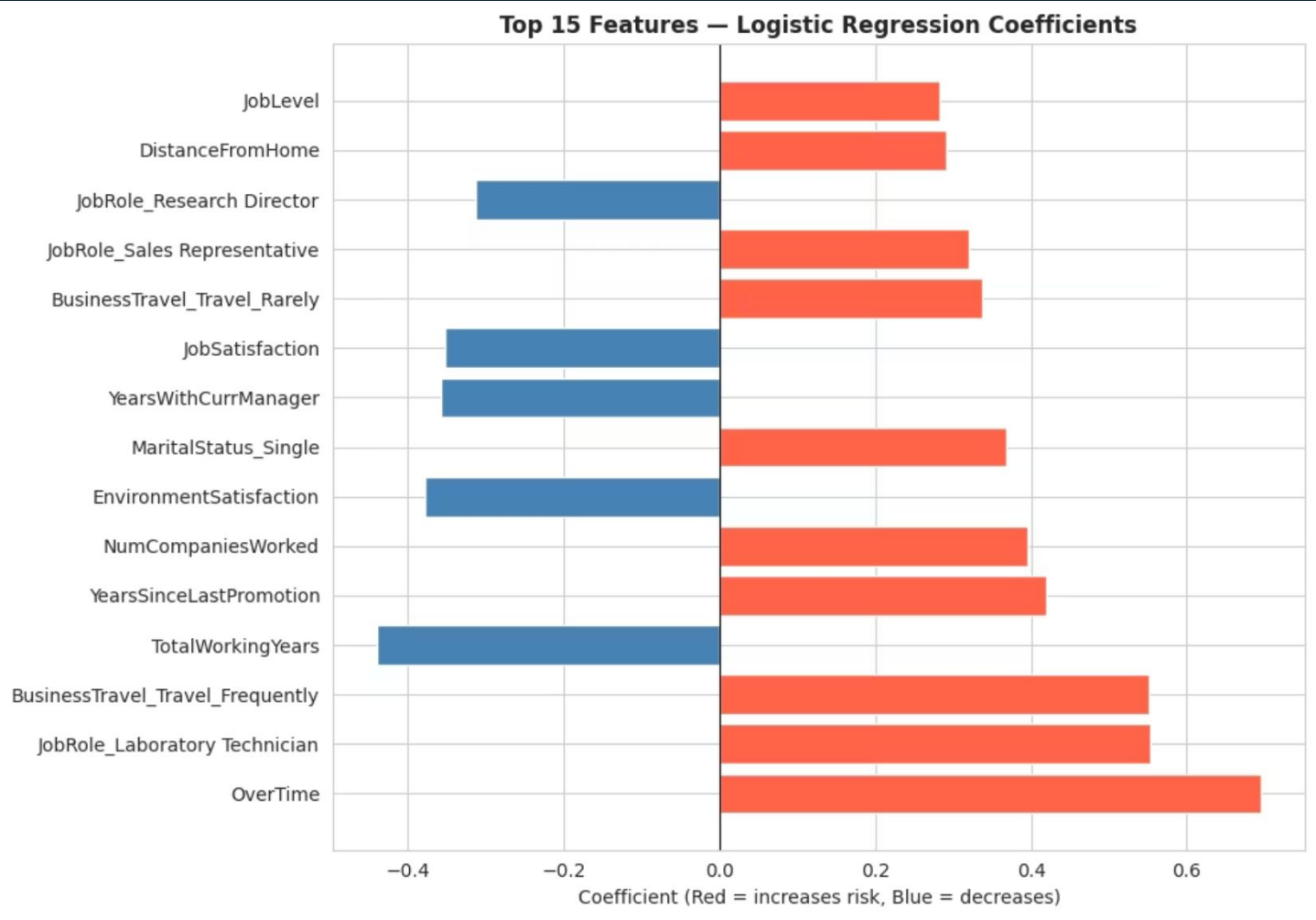
34.2% attrition rate

351 employees · 100% work overtime. Highest-risk cohort need immediate intervention.



K-Means clustering (k=3) revealed that **one-quarter of the workforce** belongs to the Burned Out segment — a structurally created risk pool.

Model Performance Summary



What the Models Tell Us

Logistic Regression (AUC 0.80) catches 66% of employees who will leave before they resign

Decision Tree (AUC 0.71) produces plain-English rules managers can act on directly

OverTime is the #1 predictor in both models are consistent and unambiguous



Both models agree: overtime status is the dominant signal. The finding is robust across methods.

Model Comparison Summary

	Technique	Accuracy	ROC-AUC	Top Predictors	Key Insight
0	Logistic Regression (Tuned)	77%	0.804100	OverTime, Travel, TotalWorkingYears	Overtime + low tenure = highest risk
1	Decision Tree (Tuned)	73%	0.712200	TotalWorkingYears, OverTime, Income	Early career + overtime + low pay = leaving

Three Actionable Recommendations



1. Reduce Mandatory Overtime

The highest-ROI intervention available. Cutting overtime exposure directly attacks the #1 attrition driver across all models.



2. Targeted Retention for High-Risk Roles

Deploy role-specific programs for **Sales Representatives (39.8%)** and **Laboratory Technicians (23.9%)**, not a one-size-fits-all policy.



3. Accelerate Early-Career Promotion Cycles

Provide visible growth pathways to retain the Early Career segment before they explore outside options.



Limitations & Closing Thoughts

Single Company Snapshot

Findings may not generalize beyond IBM's workforce structure

No Exit Interview Data

Self-reported reasons for leaving are absent from the dataset

Correlation $\neq$ Causation

Statistical associations do not confirm causal mechanisms

Bottom Line

Attrition is predictable.

With the right data and the right interventions, organizations can identify at-risk employees before they walk out the door and act decisively to retain them.

Thank You

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Dataset

IBM HR Analytics • 1,470 employees • 35 features

Methods

Logistic Regression • K-Means • Decision Tree

Key Finding

Overtime is the #1 driver and it is controllable and actionable

